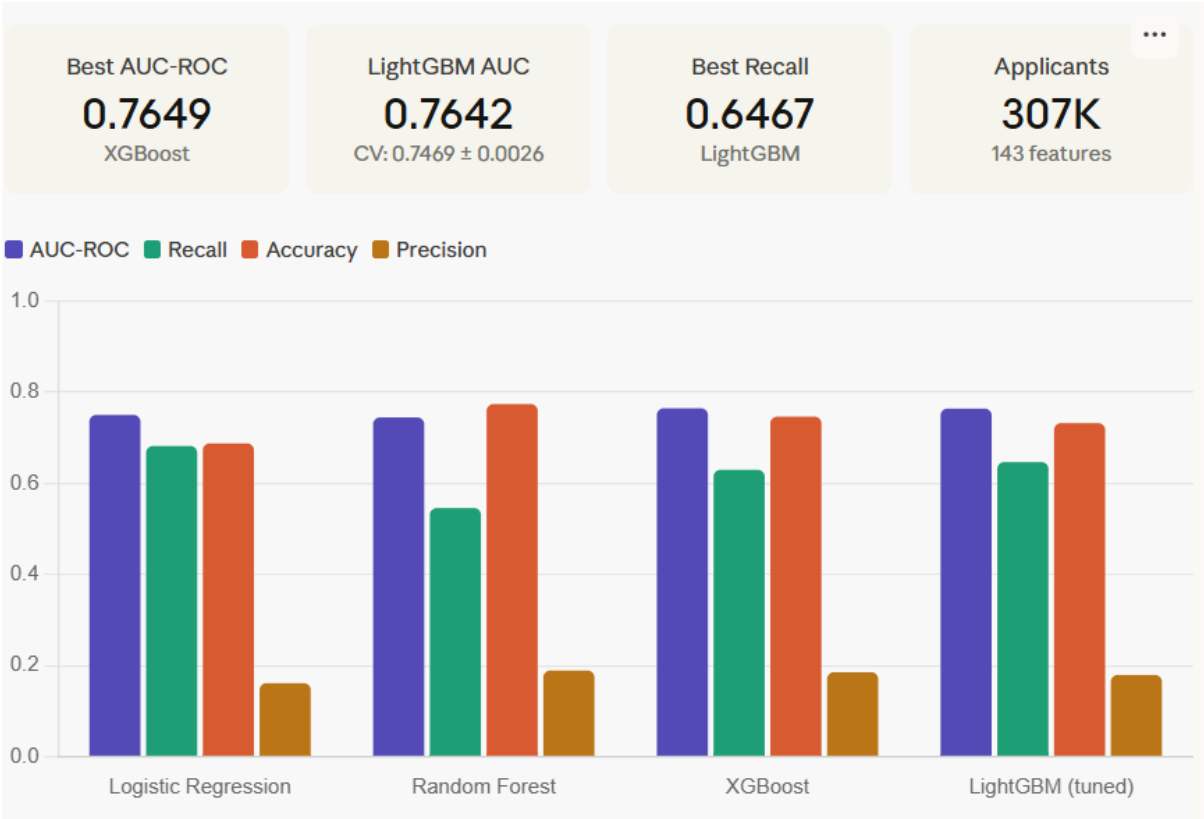
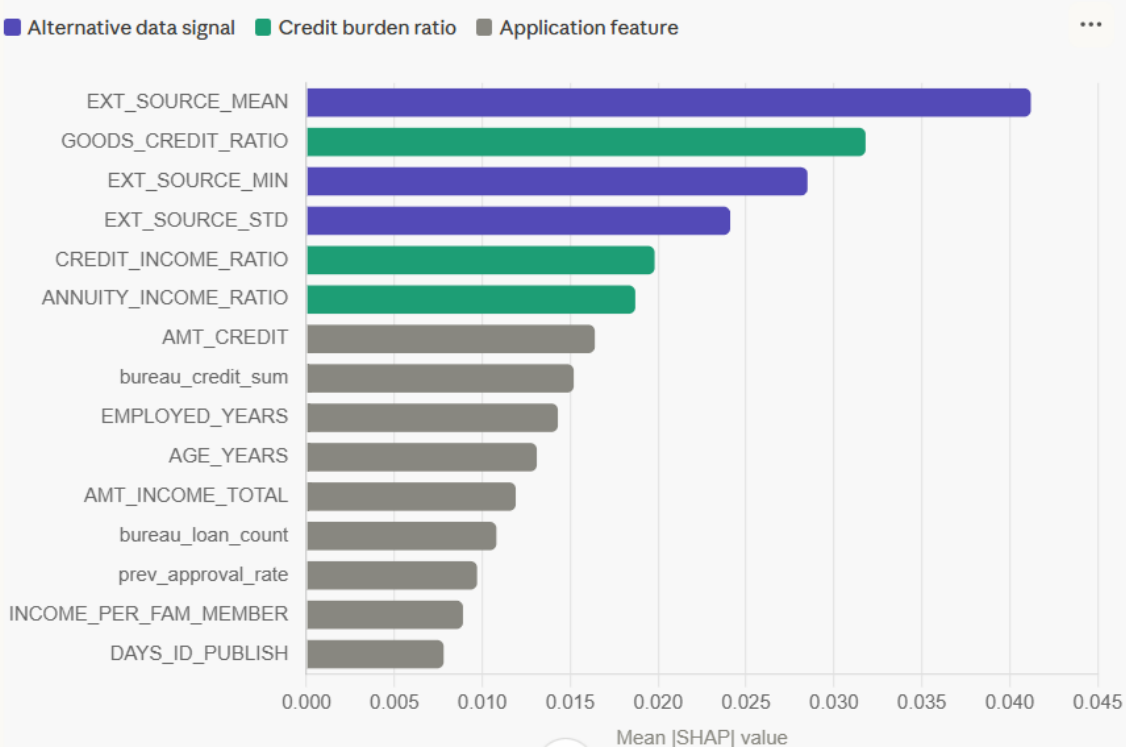
Full model comparison

Model	AUC-ROC	Accuracy	Recall
Logistic Regression	0.7503	0.6878	0.6820
Random Forest	0.7449	0.7742	0.5458
XGBoost	0.7649	0.7466	0.6298
LightGBM (tuned)	0.7642	0.7323	0.6467

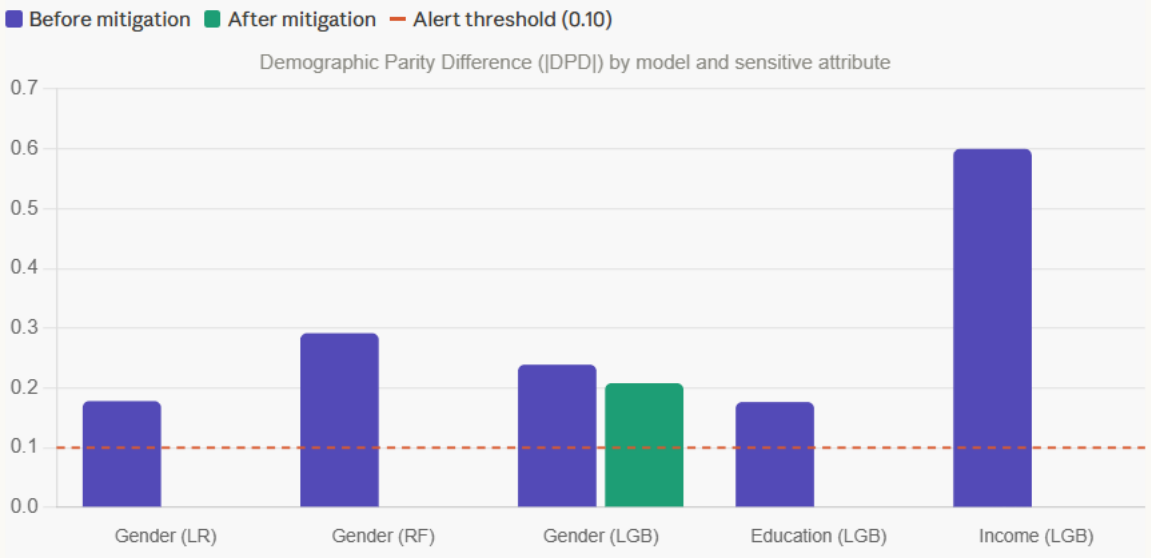
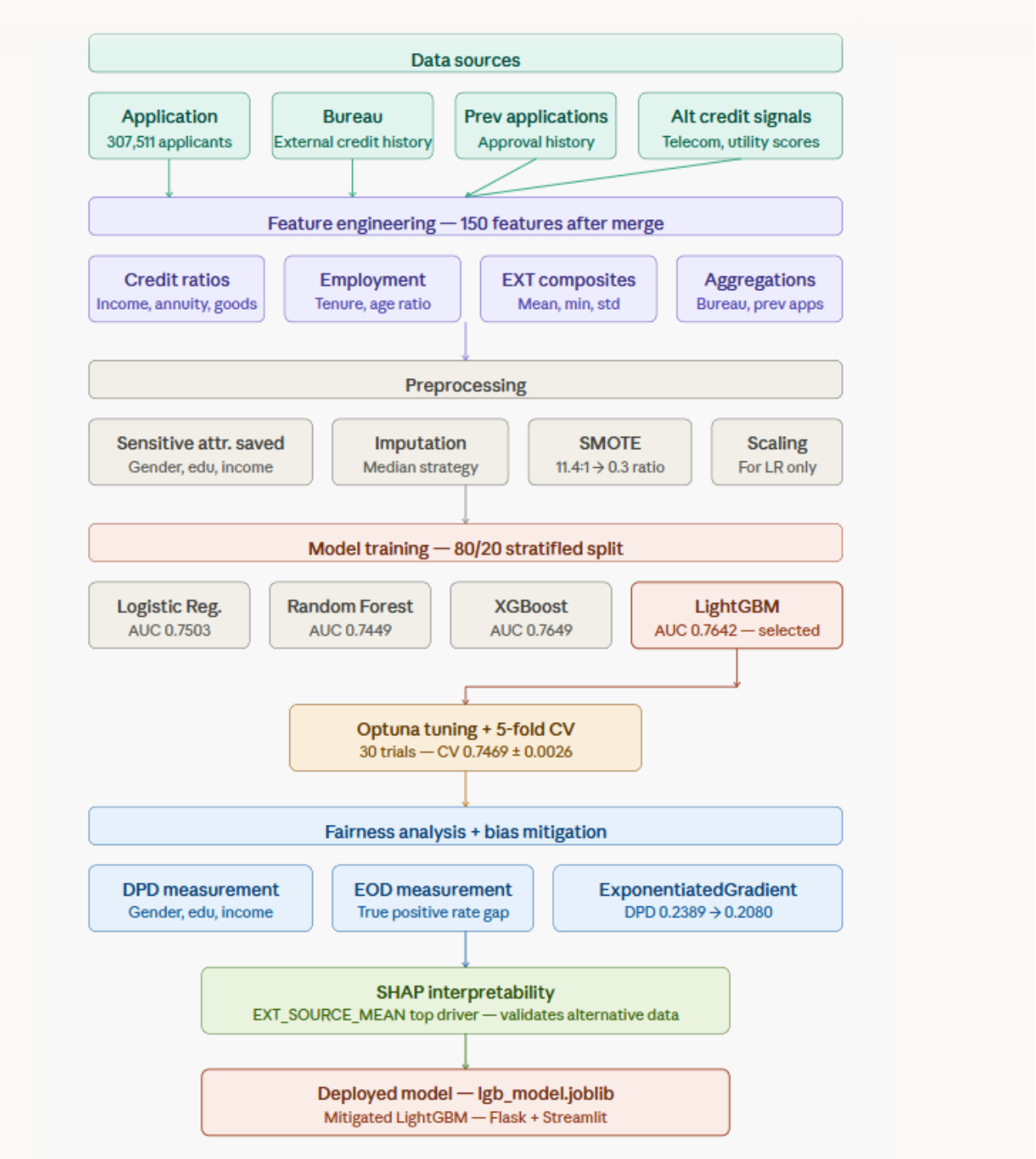


Now the SHAP feature importance — showing the alternative data signals at the top, which is the heart of the SDG argument.



Finally, the fairness impact visual — showing DPD and EOD before vs after mitigation across all sensitive attributes.

Gender DPD — before	Gender DPD — after	Gender EOD — before	Gender EOD — after
0.2389	0.2080 13% reduction	0.6923	0.6481 6.4% reduction



Full data pipeline Pipeline